

Assignment: Implement Sorting and Searching Algorithms

Sorting Algorithms

Implement the following sorting algorithms:

1. **Bubble Sort**
2. **Selection Sort**
3. **Insertion Sort**
4. **Merge Sort**
5. **Quick Sort**

Searching Algorithms

Implement the following searching algorithms:

1. **Linear Search**
2. **Binary Search**
3. **Two-Pointer Technique**

Create a problem statement of your Own that involves applying algorithms on linked lists. Aim for a medium or hard difficulty level to challenge problem-solving skills. For your custom problem:

- Define the problem clearly.
- Write the code to implement the solution.
- Analyze the time and space complexity of your solution.
- Have Distinct Problem statement for each Candidate.

Submission Guidelines

1. Raise a separate Pull Request (PR) for each algorithm.
2. In the PR description, include:
 - a. The problem statement.
 - b. Time complexity analysis.
 - c. Space complexity analysis.
3. Ensure your code follows clean code principles and adheres to best practices.

4. Explore edge cases and test your implementations thoroughly with corner cases.
5. Deadline for this Assignment would be Tuesday EOD (5:30 pm).